

Networking Fundamentals

Topic 1: IP Addressing

An IP address is a unique identifier for a device on a network. IPv4 uses 32 bits, typically formatted as four decimal numbers (e.g., 192.168.1.1). IPv6 uses 128 bits, represented in hexadecimal. IP addresses are used for routing data packets across networks.

Topic 2: DNS (Domain Name System)

DNS translates human-readable domain names (like example.com) into IP addresses that computers use to identify each other on the network. It functions essentially as the phonebook of the Internet.

Topic 3: HTTP

HTTP (Hypertext Transfer Protocol) is the foundation of data communication for the World Wide Web. It is an application layer protocol used to transmit hypermedia documents, such as HTML. It operates on a client-server model, where a client sends a request and a server responds.